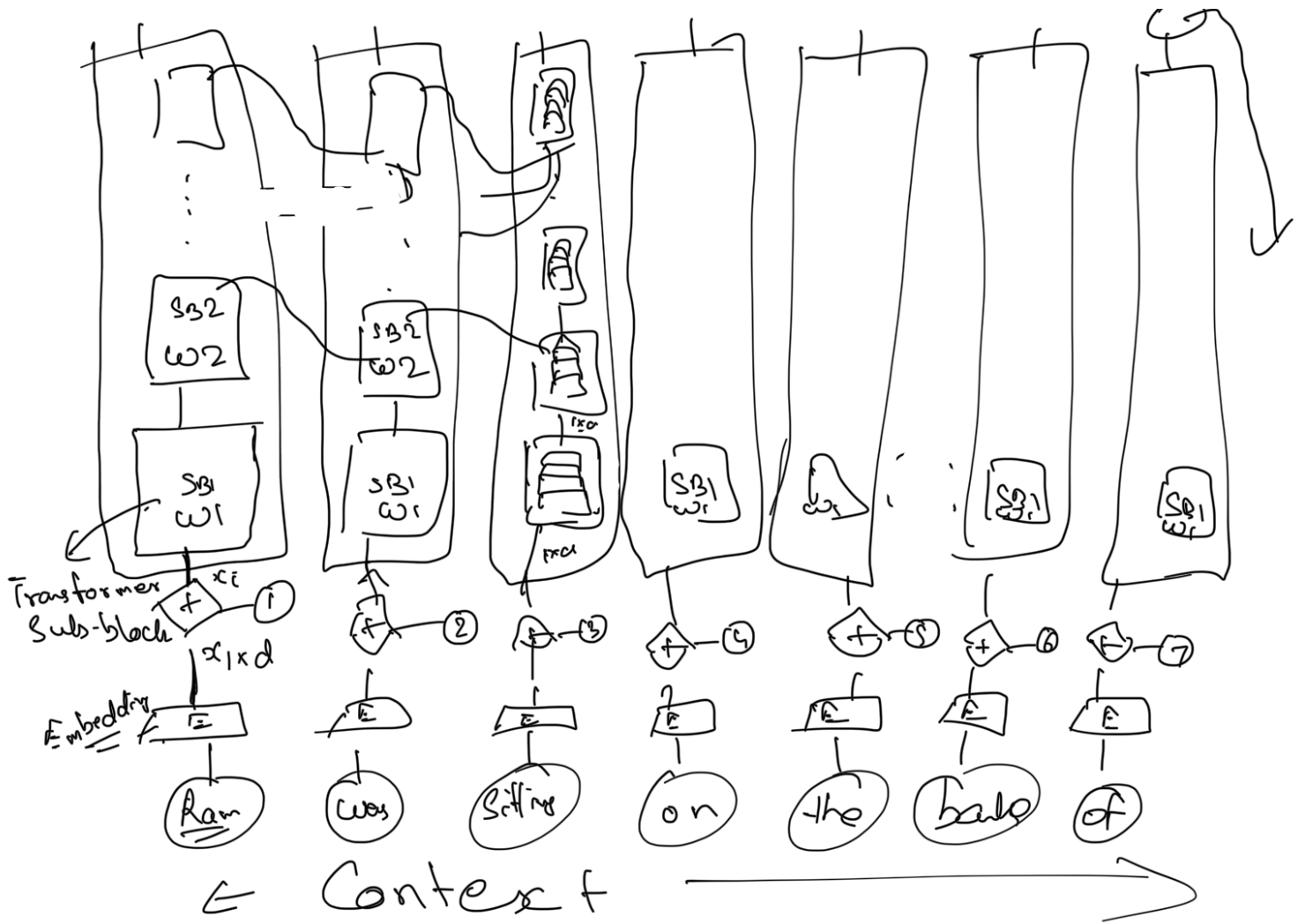
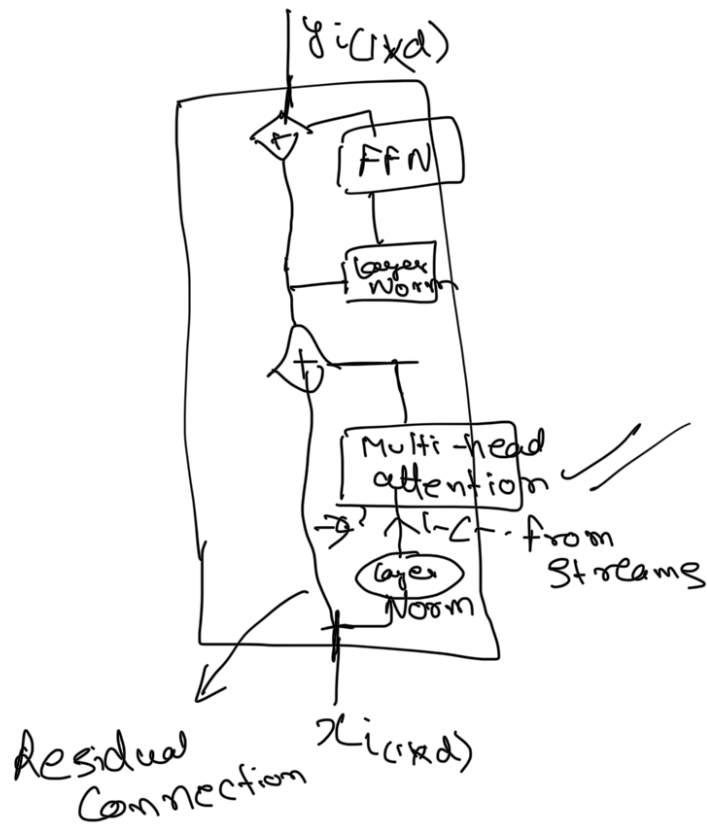




- All streams consist of sequence of transformer sub-block
- ^{Number of} Transformer sub-blocks in all stream is same.
- Weights inside transformer sub-block at ^{the} same among all streams is same.

Transformer Sub-block



Layer Norm $\Rightarrow$ Normalization across a vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

$$\hat{x} = \frac{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} - \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix}}{\sqrt{\frac{\sigma^2 + \epsilon}{d}}}$$

$\epsilon = 10^{-5}$

$$\mu = \frac{1}{d} \sum_{i=1}^d x_i$$

$$\sigma^2 = \frac{1}{d} \sum_{i=1}^d (x_i - \mu)^2$$

Batch Normalization

- 1) Operation per batch
- 2) operation done on vectors in a batch

Layer Normalization

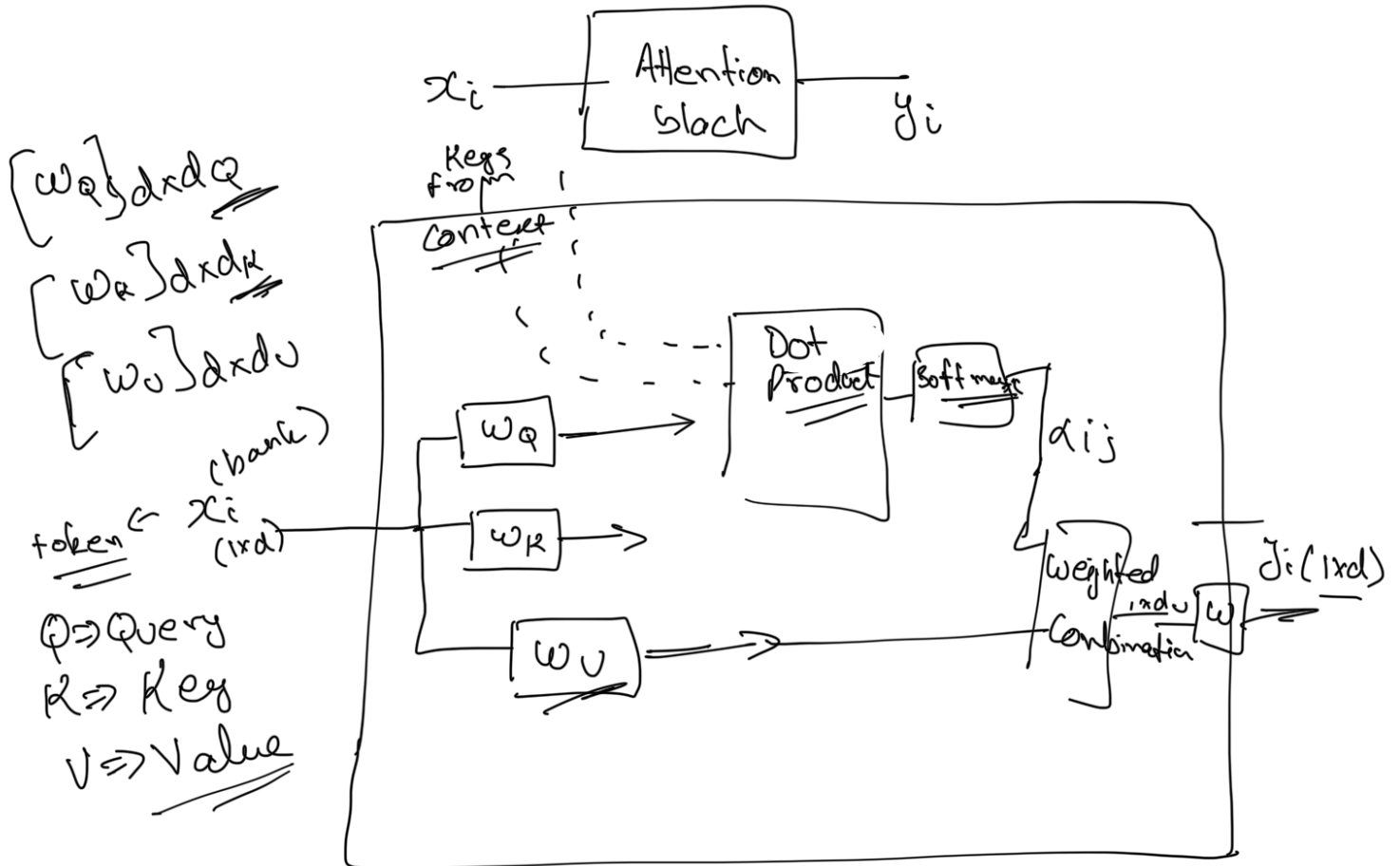
- operation per samples
- Operation done on components of a vector in a single sample

Multi-head Attention $\Rightarrow$

Combination of
many Self-Attentions.

$\begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix}$

Attention block



Query: Present token

Key: Representation with which
you compare your
present token.

Weights α_{ij} = $\text{Softmax} (q_i k_j^T)$
 (quantifying relation between two tokens)
 $i \Rightarrow$ present token
 $j \Rightarrow$ neighbouring token

$$q = (x_i)^Q = x_i W_Q$$

$$k = (x_j)^K = x_j W_K$$

$$\alpha_{ij} = \text{Softmax} \left((x_i)^Q (x_j)^K^T \right)$$

$$\alpha_{ij} = \frac{\text{Softmax} \left((x_i)^Q (x_j)^K^T \right)}{\sqrt{d}}$$

$$(x_j)^V = x_j W_V$$

$$\text{head}^i = \sum_{j=1}^N \alpha_{ij} (x_j)^V$$

$i \Rightarrow$ present index

$j \Rightarrow$ neighbouring indices

$N \Rightarrow$ Context window

~~self~~ self attention
 weighted attention

$$y_i = \text{head}^i W_{d \times d}$$

Output from attention block

dimension d

Value $\Rightarrow$ The information that
 one transformer
 sub-block will passing
 on to another
 transformer sub-block.